Exoplanets Spacing - Collatz Sequences on 3D Geometry Octave

Advanced Planetary Predictor: A Software Tool for Researchers / Exoplanetary System Architectures Using Fundamental Constants

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Abstract

The 3DCOM Advanced Planetary Predictor is a novel software application designed to test the hypothesis that planetary orbits in a system can be predicted using a geometric progression based on fundamental constants LZ (~1.23498228) and HQS (~0.235501). The tool implements a resonance model where planetary formation is favored at orbital distances where a calculated resonance strength, derived from these constants and a selectable mathematical function, exceeds a defined threshold. This paper describes the application, which provides an interactive environment for comparing predicted orbits against 14 known exoplanetary systems, tuning resonance parameters, and evaluating the predictive accuracy of progressively refined constant sets. The software offers a powerful platform for hypothesis testing, pattern recognition, and data export, facilitating further research into the fundamental ordering principles of planetary systems. For Planetary Orbital Quantization we use **a novel mathematical framework** based on **Collatz Sequences** set it up on a 3D Geometry with root numbers in Octave. Orbital distances follow a discrete scaling law:

$$a\_n = a\_0 \cdot LZ\hat{\ }n \cdot F\_HQS(n)$$

where F_HQS(n) is a resonance function (e.g., tanh, sin) modulated by HQS.
Accurately predict planetary orbits in Solar System, TRAPPIST-1, and Kepler-90...etc
In this paper **I present data** for 14 Systems. Data extracted from app
Logos_Astro_xl_app.py or
Logos_Astro_xl_app V.4.2.0, available in github.

Introduction

1. 3DCOM is a novel framework where reality emerges as a discrete, computational substrate governed by a 3D Collatz - octave operator (**the Logos**).

$$\hat{\Lambda} \circ v(\vec{n}) = \begin{cases} \frac{v(\vec{n})}{2} & \text{if } v(\vec{n}) \equiv 0 \pmod 2 \\ 3 \cdot v(\vec{n}) + \Pi(\vec{n}) & \text{if } v(\vec{n}) \equiv 1 \pmod 2 \end{cases}$$

2. The LZ constant (Loop Zero)= 1.234928228.

$$\Psi^* = \mathcal{F}(\Psi^*) = \sin(\Psi^*) + e^{-\Psi^*}$$

This transcendental equation is solved numerically. The solution is the LZ constant:

$$\Lambda_{LZ} = \Psi^* \approx 1.23498228$$

Thus, **the LZ constant is the attractor of the recursive wave function**: $\Lambda_{LZ} = \Psi^*$.

3. HQS (Harmonic Quantum Shift)= 0.2355012867

HQS is the curvature of the emergent spacetime, derived from the energy cost of recursion.

3A: Definition of the Recursive Energy Cost

The change in total activity at each step is the "effort" required for computation:

$$\Delta\Psi(k) = |\Psi(k) - \Psi(k-1)| = |\mathcal{F}(\Psi(k-1)) - \Psi(k-1)|$$

The cumulative cost up to step k is:

$$\text{HQS}_{\text{cumulative}}(k) \propto \sum_{i=1}^{k} \Delta\Psi(i)$$

3B: Identifying the Per-Step Binding Energy

At each step, the exponential term $e^{-\Psi(k-1)}$ acts as a **binding energy,** preventing collapse and enforcing structure. The fraction of the previous state's value used for this is:

$$f_{\text{bind}}(k) = \frac{e^{-\Psi(k-1)}}{\Psi(k-1)}$$

3C: The Equilibrium HQS Constant

At the fixed point Ψ^*, this binding fraction becomes a constant. This is the HQS:

$$\text{HQS} = \lim_{k\to\infty} f_{\text{bind}}(k) = \frac{e^{-\Psi^*}}{\Psi^*}$$

Substituting the value of Ψ^*:

$$\text{HQS} = \frac{e^{-1.23498228}}{1.23498228} \approx \frac{0.2905}{1.23498} \approx 0.235$$

Therefore, HQS is approximately 0.235, or 23.5%.

HQS is the immutable fraction of a node's computational energy that must be dedicated to maintaining its structural integrity within the emergent spacetime. It is the energy "tax" paid at each recursion step to create a stable, curved reality. It is directly interpreted as the source of the emergent Ricci curvature (R):

$$R(\vec{n}) \propto \text{HQS} \cdot v(\vec{n})$$

There are **5 values before LZ stabilize,** so **we use all of theme for research**.

Lz_1 = 1.20935043	HQS = 0.246736624	x = 6.408896718770941	a= 0.072973525643
Lz_2 = 1.23377754	HQS = 0.2360154114	x = 5.5873802919838305	a= 0.07297352564300001
Lz_3 = 1.23493518	HQS = 0.2355213614	x = 5.552617415444504	a= 0.072973525643
Lz_4 = 1.23498046	HQS = 0.2355020624	x = 5.551264532511617	a= 0.072973525643
Lz_5 = 1.23498221	HQS = 0.2355013165	x = 5.551212254753354	a= 0.07297352564299998
LZ _s = 1.23498228	HQS = 0.2355012867	x = 5.551210763930036	a= 0.07297351640903436

Logos_Astro_xl_app

Methodology:

The application is built in Python using the Tkinter GUI framework. The core predictive algorithm calculates a sequence of orbital distances `a_n` for a continuous index `n` using the relationship $a_n = a_0 \times (LZ)^n$, where a_0 is the innermost planet's semi-major axis. A resonance function `R(n) = HQS × f(n, param)`, where `f` is a user-defined function (e.g., sinusoidal, hyperbolic tangent), is evaluated at each `n`. Orbits where `|R(n)|` exceeds a tunable threshold are flagged as stable and predicted to host a planet. The model's performance is quantitatively assessed by its ability to retrospectively "predict" the known orbits of documented exoplanetary systems within a specified tolerance. The software features a graphical interface for parameter manipulation, visual validation through integrated plots, and tools for batch comparison and data export, ensuring reproducibility and extensibility of the research.

This is the concept: no vacuum, planetary orbits are on stable resonance.

The user version you find it in github with a py code and a dist file.

Bellow data from the app with parameters.

Systems planetary spacing data

1.
=== Trappist-1 PREDICTED PLANETARY ORBITS ===

Using constants: LZ

LZ: 1.2349822879948562

HQS: 0.2355012833

Resonance function: **exp(0.08)**

n range: 0.0 to 24.0 (step: 0.001)

Predicted orbits: 13497 found

0.011540 AU

0.011542 AU

0.011545 AU

0.011547 AU

0.011550 AU

0.011552 AU

0.011555 AU

0.011557 AU

0.011560 AU

0.011562 AU

... and 13487 more

Actual orbits: [0.01154, 0.0158, 0.02227, 0.02925, 0.03849, 0.04683, 0.06189, 0.081]

Prediction accuracy: 8/8 planets predicted correctly

Matched orbits (actual, predicted):

0.01154 AU -> 0.011540 AU
0.0158 AU -> 0.015645 AU
0.02227 AU -> 0.022051 AU
0.02925 AU -> 0.028963 AU
0.03849 AU -> 0.038107 AU
0.04683 AU -> 0.046362 AU
0.06189 AU -> 0.061282 AU
0.081 AU -> 0.080205 AU

2.

=== Solar System PREDICTED PLANETARY ORBITS ===

=== Solar System PREDICTED PLANETARY ORBITS ===

Using constants: LZ
LZ: 1.2349822879948562
HQS: 0.2355012833
Resonance function: **exp(0.05)**
n range: 0.0 to 24.0 (step: 0.001)
Predicted orbits: 24000 found

0.387000 AU
0.387082 AU
0.387163 AU
0.387245 AU
0.387327 AU
0.387409 AU
0.387490 AU
0.387572 AU
0.387654 AU
0.387736 AU
... and 23990 more

Actual orbits: [0.387, 0.723, 1.0, 1.524, 5.203, 9.539, 19.18, 30.06]

Prediction accuracy: 8/8 planets predicted correctly

Matched orbits (actual, predicted):

0.387 AU -> 0.387000 AU
0.723 AU -> 0.715831 AU
1.0 AU -> 0.990131 AU
1.524 AU -> 1.508856 AU
5.203 AU -> 5.151454 AU
9.539 AU -> 9.444510 AU
19.18 AU -> 18.988243 AU
30.06 AU -> 29.759831 AU
0.723 AU -> 0.715831 AU
1.0 AU -> 0.990131 AU
1.524 AU -> 1.508856 AU
5.203 AU -> 5.151454 AU
9.539 AU -> 9.444510 AU
19.18 AU -> 18.988243 AU
30.06 AU -> 29.759831 AU

3.

=== Kepler-90 PREDICTED PLANETARY ORBITS ===

Using constants: LZ

LZ: 1.2349822879948562

HQS: 0.2355012833

Resonance function: **exp(0.07)**

n range: 0.0 to 24.0 (step: 0.001)

Predicted orbits: 17332 found

0.074000 AU
0.074016 AU
0.074031 AU
0.074047 AU
0.074062 AU
0.074078 AU
0.074094 AU
0.074109 AU
0.074125 AU

0.074141 AU
... and 17322 more

Actual orbits: [0.074, 0.089, 0.107, 0.32, 0.42, 0.48, 0.71, 1.01]

Prediction accuracy: 8/8 planets predicted correctly (First planet included)
Matched orbits (actual, predicted):
0.074 AU -> 0.074000 AU
0.089 AU -> 0.088112 AU
0.107 AU -> 0.105939 AU
0.32 AU -> 0.316859 AU
0.42 AU -> 0.415839 AU
0.48 AU -> 0.475276 AU
0.71 AU -> 0.703035 AU
1.01 AU -> 0.999904 AU

4.

=== Proxima Centauri PREDICTED PLANETARY ORBITS ===

Using constants: LZ
LZ: 1.2349822879948562
HQS: 0.2355012833
Resonance function: **exp(0.06)**
n range: 0.0 to 24.0 (step: 0.001)
Predicted orbits: 22790 found
0.028100 AU
0.028106 AU
0.028112 AU
0.028118 AU
0.028124 AU
0.028130 AU
0.028136 AU
0.028142 AU
0.028147 AU
0.028153 AU
... and 22780 more

Actual orbits: [0.0281, 0.04848, 1.489]

Prediction accuracy: 3/3 planets predicted correctly

Matched orbits (actual, predicted):

0.0281 AU -> 0.028100 AU

0.04848 AU -> 0.048001 AU

1.489 AU -> 1.474133 AU

5.

=== HD 10180 PREDICTED PLANETARY ORBITS ===

Using constants: LZ

LZ: 1.2349822879948562

HQS: 0.2355012833

Resonance function: **exp(0.02)**

n range: 0.0 to 24.0 (step: 0.001)

Predicted orbits: 24000 found

0.022200 AU

0.022205 AU

0.022209 AU

0.022214 AU

0.022219 AU

0.022223 AU

0.022228 AU

0.022233 AU

0.022238 AU

0.022242 AU

... and 23990 more

Actual orbits: [0.0222, 0.06412, 0.0904, 0.12859, 0.2699, 0.33, 0.4929, 1.427, 3.381]

Prediction accuracy: 9/9 planets predicted correctly

Matched orbits (actual, predicted):

0.0222 AU -> 0.022200 AU

0.06412 AU -> 0.063480 AU

0.0904 AU -> 0.089508 AU

0.12859 AU -> 0.127305 AU

0.2699 AU -> 0.267208 AU
0.33 AU -> 0.326740 AU
0.4929 AU -> 0.488033 AU
1.427 AU -> 1.413007 AU
3.381 AU -> 3.347818 AU

6.

=== Kepler-11 PREDICTED PLANETARY ORBITS ===

Using constants: LZ
LZ: 1.2349822879948562
HQS: 0.2355012833
Resonance function: **exp(0.07)**
n range: 0.0 to 24.0 (step: 0.001)
Predicted orbits: 17332 found
0.091000 AU
0.091019 AU
0.091038 AU
0.091058 AU
0.091077 AU
0.091096 AU
0.091115 AU
0.091135 AU
0.091154 AU
0.091173 AU
... and 17322 more

Actual orbits: [0.091, 0.107, 0.155, 0.195, 0.25, 0.466]

Prediction accuracy: 6/6 planets predicted correctly
Matched orbits (actual, predicted):
0.091 AU -> 0.091000 AU
0.107 AU -> 0.105934 AU
0.155 AU -> 0.153459 AU
0.195 AU -> 0.193072 AU
0.25 AU -> 0.247517 AU
0.466 AU -> 0.461419 AU

7.

=== L 98-59 PREDICTED PLANETARY ORBITS ===

Using constants: LZ

LZ: 1.2349822879948562

HQS: 0.2355012833

Resonance function: **exp(0.02)**

n range: 0.0 to 24.0 (step: 0.001)

Predicted orbits: 24000 found

0.018800 AU
0.018804 AU
0.018808 AU
0.018812 AU
0.018816 AU
0.018820 AU
0.018824 AU
0.018828 AU
0.018832 AU
0.018836 AU
... and 23990 more

Actual orbits: [0.0188, 0.0223, 0.0309, 0.0494, 0.0712, 1.052]

Prediction accuracy: 6/6 planets predicted correctly

Matched orbits (actual, predicted):

0.0188 AU -> 0.018800 AU
0.0223 AU -> 0.022080 AU
0.0309 AU -> 0.030593 AU
0.0494 AU -> 0.048908 AU
0.0712 AU -> 0.070492 AU
1.052 AU -> 1.041665 AU

8.

=== K2-18 PREDICTED PLANETARY ORBITS ===

Using constants: LZ

LZ: 1.2349822879948562

HQS: 0.2355012833

Resonance function: **exp(0.02)**

n range: 0.0 to 24.0 (step: 0.001)

Predicted orbits: 24000 found

0.067000 AU
0.067014 AU
0.067028 AU
0.067042 AU
0.067057 AU
0.067071 AU
0.067085 AU
0.067099 AU
0.067113 AU
0.067127 AU
... and 23990 more

Actual orbits: [0.067, 0.1591]

Prediction accuracy: 2/2 planets predicted correctly

Matched orbits (actual, predicted):

0.067 AU -> 0.067000 AU
0.1591 AU -> 0.157541 AU

9.

=== HD 101581 PREDICTED PLANETARY ORBITS ===

Using constants: LZ

LZ: 1.2349822879948562

HQS: 0.2355012833

Resonance function: **exp(0.02)**

n range: 0.0 to 24.0 (step: 0.001)

Predicted orbits: 24000 found

0.046000 AU
0.046010 AU
0.046019 AU
0.046029 AU
0.046039 AU

0.046049 AU
0.046058 AU
0.046068 AU
0.046078 AU
0.046087 AU
... and 23990 more

Actual orbits: [0.046, 0.0573, 0.0671]

Prediction accuracy: 3/3 planets predicted correctly

Matched orbits (actual, predicted):
0.046 AU -> 0.046000 AU
0.0573 AU -> 0.056737 AU
0.0671 AU -> 0.066440 AU

10.

=== HD 73344 PREDICTED PLANETARY ORBITS ===

=== HD 73344 PREDICTED PLANETARY ORBITS ===

Using constants: LZ
LZ: 1.2349822879948562
HQS: 0.2355012833
Resonance function: **exp(0.02)**
n range: 0.0 to 24.0 (step: 0.001)
Predicted orbits: 24000 found
0.131000 AU
0.131028 AU
0.131055 AU
0.131083 AU
0.131111 AU
0.131138 AU
0.131166 AU
0.131194 AU
0.131221 AU
0.131249 AU
... and 23990 more

Actual orbits: [0.131, 0.343, 6.7]

Prediction accuracy: 3/3 planets predicted correctly

Matched orbits (actual, predicted):

0.131 AU -> 0.131000 AU
0.343 AU -> 0.339575 AU
6.7 AU -> 6.634286 AU

11.

=== Barnard Star PREDICTED PLANETARY ORBITS ===

Using constants: LZ
LZ: 1.2349822879948562
HQS: 0.2355012833
Resonance function: **exp(0.02)**
n range: 0.0 to 24.0 (step: 0.001)
Predicted orbits: 24000 found

0.018800 AU
0.018804 AU
0.018808 AU
0.018812 AU
0.018816 AU
0.018820 AU
0.018824 AU
0.018828 AU
0.018832 AU
0.018836 AU
... and 23990 more

Actual orbits: [0.0188, 0.0229, 0.0274, 0.0381]

Prediction accuracy: 4/4 planets predicted correctly

Matched orbits (actual, predicted):

0.0188 AU -> 0.018800 AU
0.0229 AU -> 0.022675 AU
0.0274 AU -> 0.027131 AU
0.0381 AU -> 0.037726 AU

12.

=== WASP-47 PREDICTED PLANETARY ORBITS ===

Using constants: LZ
LZ: 1.2349822879948562
HQS: 0.2355012833
Resonance function: **exp(0.02)**
n range: 0.0 to 24.0 (step: 0.001)
Predicted orbits: 24000 found
  0.017000 AU
  0.017004 AU
  0.017007 AU
  0.017011 AU
  0.017014 AU
  0.017018 AU
  0.017022 AU
  0.017025 AU
  0.017029 AU
  0.017032 AU
  ... and 23990 more

Actual orbits: [0.017, 0.052, 0.087, 1.42]

Prediction accuracy: 4/4 planets predicted correctly
Matched orbits (actual, predicted):
  0.017 AU -> 0.017000 AU
  0.052 AU -> 0.051483 AU
  0.087 AU -> 0.086145 AU
1.42 AU -> 1.406020 AU

13.

=== Gliese 667 PREDICTED PLANETARY ORBITS ===

Using constants: LZ
LZ: 1.2349822879948562

HQS: 0.2355012833

Resonance function: **exp(0.02)**

n range: 0.0 to 24.0 (step: 0.001)

Predicted orbits: 24000 found

0.050431 AU
0.050442 AU
0.050452 AU
0.050463 AU
0.050474 AU
0.050484 AU
0.050495 AU
0.050506 AU
0.050516 AU
0.050527 AU
... and 23990 more

Actual orbits: [0.050431, 0.12501, 0.25]

Prediction accuracy: 3/3 planets predicted correctly

Matched orbits (actual, predicted):

0.050431 AU -> 0.050431 AU
0.12501 AU -> 0.123772 AU
0.25 AU -> 0.247535 AU

14.

=== HD 209458 PREDICTED PLANETARY ORBITS ===

Using constants: LZ

LZ: 1.2349822879948562

HQS: 0.2355012833

Resonance function: **exp(0.02)**

n range: 0.0 to 24.0 (step: 0.001)

Predicted orbits: 24000 found

0.047070 AU
0.047080 AU
0.047090 AU

0.047100 AU
0.047110 AU
0.047120 AU
0.047130 AU
0.047140 AU
0.047150 AU
0.047159 AU
... and 23990 more

Actual orbits: [0.04707]

Prediction accuracy: 1/1 planets predicted correctly
Matched orbits (actual, predicted):
0.04707 AU -> 0.047070 AU

Reference:

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